

Article

AHP and GCA Combined Approach to Green Design Evaluation of Kindergarten Furniture

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Abstract: Kindergarten furniture is an important part of children's furniture. However, in sharp contrast with the high level of concern about the color, shape, structure, and modularity of kindergarten furniture, the research on green design and green evaluation of kindergarten furniture has not been given due attention. Through understanding the concept and principles of green design, this paper presents an objective approach to evaluate the green design of kindergarten furniture. An evaluation method based on the analytic hierarchy process (AHP) integrating gray correlation analysis (GCA) for kindergarten furniture green design was proposed. By using the AHP to determine the green design elements of kindergarten furniture, five standard layer indicators were obtained in aspects of environmental friendliness of the materials, color, technological structure, size, and interestingness, as well as 11 criterion layer indicators: environmental friendliness of structural materials, environmental friendliness of adhesives, environmental friendliness of paint, stimulate creativity or enhance concentration, avoid the feelings of unsteadiness, structural stability, rounding the edges, comfortable to use, size adjustable, interesting decorative patterns, and modular. Afterward, GCA was used to evaluate the green design of five kindergarten furniture schemes and a comprehensive evaluation score was obtained. Taking the design scheme represented by five desk and chair sets in kindergarten as an example, the effectiveness of this method was verified, and the advantages and disadvantages of the five desk and chair sets were analyzed and compared. The results showed that the third desk and chair sets at kindergarten made of New Zealand pine ranks first in the comprehensive green evaluation indicators, followed by the fourth sets made of rubber and wood. After the green design evaluation, we studied the carbon footprint of five furniture products and analyzed their carbon emissions at the production stage, the packaging stage, the transportation stage, and the storage stage. The results show that the amount of carbon emission is generally consistent with the evaluation results of the green scheme, which verifies the effectiveness of the green design evaluation method. This study provides an effective and feasible reference for kindergarten furniture designers. Hence, a greener kindergarten furniture design is expected to improve the learning and living environment for kindergarten-age children.

Keywords: kindergarten furniture; green design; analytic hierarchy process (AHP); gray correlation analysis (GCA)



Citation: Xie, X.; Zhu, J.; Ding, S.; Chen, J. AHP and GCA Combined Approach to Green Design Evaluation of Kindergarten Furniture. *Sustainability* **2024**, *16*, 1. <https://doi.org/10.3390/su16010001>

Academic Editor: Steve Kardinal Jusuf

Received: 11 October 2023

Revised: 1 December 2023

Accepted: 15 December 2023

Published: 19 December 2023

Corrected: 1 November 2024



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1. Introduction

Kindergarten is an important place for children's activities, learning, and life, and early childhood is a key basic period of life. In early childhood, young children's organs and body tissues are not mature and have strong plasticity in various aspects such as personality, intelligence, and physical development. However, they are greatly affected by

the environment, and the quality of their environment directly affects their physical and mental health and development. The application of green design in kindergarten furniture can create a good living and learning environment for young children, meet the needs of teachers, strive to enable young children to obtain a good education, and enable them to fully develop their intelligence, personality, body, and psychology. Therefore, it is necessary to explore the green design in kindergarten furniture.

Green design is also known as environmental design and environmental awareness design. During the entire life cycle of a product, it is important to consider the environmental attributes of the product (disassemblability, recyclability, maintainability, reusability, etc.) and consider them as design objectives. While meeting the environmental objectives, it is also necessary to ensure that the product meets the functional, service life, quality, and other requirements. The principle of green design is widely recognized as the “3R” principle, namely, “Reduce, Reuse, Recycle”—reducing environmental pollution, reducing energy consumption, and recycling or reusing products and components [1].

The green design emphasizes the satisfaction of the overall space in terms of environmental friendliness, energy conservation, safety, health, convenience, comfort, etc., such as indoor layout, spatial scale, decorative materials, lighting conditions, color configuration, etc., which can meet the physical, psychological, hygienic, safety, health, and other requirements of contemporary society. The basic idea is to incorporate environmental factors and pollution prevention measures into the entire design process at the design stage, taking environmental performance as the design goal and starting point, and striving to minimize the impact of design results on the environment.

This paper is organized as follows: Section 1 describes the introduction, Section 2 reviews the literature on green furniture design, the design of kindergarten furniture, and the evaluation of green design. Section 3 describes the proposed evaluation methods and the establishment of an evaluation index system. Section 4 describes the results of a case study design of five kindergarten furniture design schemes, and a discussion on the findings. Section 5 derives some conclusions and further research topics.

2. Literature Review

2.1. Green Furniture Design

The green design adheres to the principle of putting people first, seeking harmony between people and the overall spatial environment, and always based on human behavior, physiology, and psychological characteristics in the design, striving to meet the needs of human beings and achieve optimal benefits in the indoor environment. Green design requires green, environmental friendliness, and energy conservation, mainly referring to the rational and effective use of natural energy. The requirements are beneficial to health, mainly including a non-toxic, pollution-free, and fireproof living environment. It is required to meet the spiritual needs of indoor people, mainly through the appeal of art in design to enable people to obtain spiritual enjoyment [2].

Green furniture has the characteristics of environmental compatibility, prolonging service life, and reflecting innovative ideas [3]. The green design concept can be integrated into the whole process of a product life cycle, from design, manufacturing, use, recycling, and other aspects of implementation one by one, improve the level of furniture design and manufacturing and transform the development of non-green furniture industry, better meet people's consumption demand for green furniture, and achieve a harmonious coexistence between man and nature and the environment.

From the perspective of furniture enterprises, they focus on reflecting their innovative capabilities in green furniture production, thereby improving their competitiveness and obtaining more benefits in the market environment. Bumgardner, M.S. et al. [4] considered emerging and innovative wood design strategies (e.g., bionics) as well as topics that have gained traction in recent years (i.e., green supply chain management and environmental labeling and related marketing communications) to improve the company's competitiveness. Sellitto, M.A. et al. [5] defined how green innovation supports the competitive advantage of an industrial furniture

cluster located in southern Brazil by investigating 245 furniture companies in the industrial cluster. The conclusions drawn are green innovation focused on operation and process does not positively influence competitive enablers but influences competitive advantages; green innovation focused on product and customers, and eco-efficiency positively influences the competitive enablers; competitive enablers based only on product and customers and on eco-efficiency positively influences the competitive advantage.

From the perspective of consumers, they focus on the safety and environmental friendliness of furniture. They hope that furniture materials are non-toxic and harmless, and on this basis, consumers will consider aspects such as the design style and symbols of furniture. Barbaritano, M. et al. [6] have investigated how environmental concern affects the relationship between design attributes and purchasing intention. This study concluded that when consumers are highly concerned about environmental issues, they are often more influenced by the symbolic dimension of design. Kwangsawat, K. et al. [7] studied consumers' demand for environmentally friendly furniture, and the results showed that consumers prefer modern and Western-style furniture, using natural materials for production, high production technology, and cool and moderate furniture.

From the perspective of sustainable design, scholars have optimized the furniture production chain from the perspectives of LCA (Life Cycle Assessment), new environmentally friendly materials, and the decrement principle. Kwangsawat, K. et al. [8] analyzed the differential carbon footprint of each furniture type based on its product life cycle to determine the criteria for selecting low-impact materials for use in desk furniture design. Wang, Y. et al. [9] obtained a waste textile-starch composite material that combines waste textiles, starch, and other components through the use of microwave expansion technology. The material is biodegradable, environmentally friendly, and non-pollution. This study combines sustainable design with composite material manufacturing, effectively solving the problems of textile waste pollution and furniture resources. Wang, Q.W. et al. [10] introduced the properties of wood plastic materials and their applications in furniture manufacturing technology. It was explained that the environmental advantages of wood plastic materials in the future should be fully utilized throughout the entire lifecycle of furniture manufacturing, sales, use, and recycling. Wu, W. et al. [11] discussed the application of the decrement principle in the design of modern mortise and tenon structures under the background of green design, aiming at exploring new ideas of modern mortise and tenon structure design.

2.2. Kindergarten Furniture Design

Kindergarten furniture has a variety of types, and its distribution is mainly determined by the spatial functions of the kindergarten. Kindertartens both at home and abroad include functional spaces such as natural areas, manual areas, language areas, science areas, book reading areas, art areas, music areas, structural activity areas, and social activity areas, but there are significant differences in their spatial patterns.

Currently, most domestic kindertartens continue to use a spatial pattern that integrates activity rooms, bedrooms, bathrooms, and clothing storage rooms, emphasizing the independence of each class to ensure that they do not interfere with each other and do not interact with each other. With the progress and development of children's physiology and psychology, it has been found that most of their thinking is obtained through direct action and perception, while this fixed and rigid spatial pattern ignores children's physiological and psychological characteristics, often limiting the development of children's personality, impeding the development of intelligence and children's communicative abilities. Furniture should provide children with forms that adapt to their needs and preferences [12]. Different from the domestic teaching model, foreign kindertartens have separate activity rooms in each class. The corners of their activity rooms are rich in content and diverse in activities. The toilets, wash basins, and cloakrooms are collectively shared, and the original corridor route has been transformed into a multi-functional hall and greenhouse space for

collective activities, changing the traditional rigid pattern and becoming a comprehensive activity space for children to communicate, play, and interact.

Current research on the design of kindergarten furniture has mainly focused on the impact of the size and function of kindergarten furniture design on the learning ability and health of preschool children. Iliev, B. et al. [13] proposed suggestions to better adapt the chair size to children's body size by measuring the body size of preschool children and comparing it with the chair size of kindergartens in three regions. Giraldi, L. et al. [14] optimized furniture, products, and graphics in preschool environments from the perspective of real attitudes, emotions, and abilities in children's lives, and proposed a set of good practices suitable for kindergarten environments, effectively improving children's educational experience. Gimenez, R. et al. [15] investigated the role of two different layouts of school furniture in the pattern legibility and spatial-temporal parameters of graphic skill acquisition. Research has found that adjustable desks facilitate the acquisition of clear graphic patterns and can enhance children's handwriting and drawing skills.

2.3. Green Design Evaluation

The life cycle assessment (LCA) is a systematic method for assessing the potential environmental impact of a product, process, or activity throughout its life cycle from resource collection, manufacturing, and use stage to disposal. It aims to comprehensively analyze and evaluate the environmental performance of different products or services to help make more sustainable decisions [16]. Bianco, I. et al. [17] have developed a life cycle assessment (LCA)-based tool that considers the main materials and processes typically used in the furniture sector. The tool has made it possible to quantify the environmental impacts of the armchair and the evaluation of four possible scenarios to enhance its environmental sustainability. This work can therefore guide the actors in furniture value chains as to the choice of the criteria able to maximize furniture sustainability throughout its life cycle. Mohd Azman, M.A.H., et al. [18] have performed a cradle-to-gate life cycle assessment (LCA) of particleboard production by using OpenLCA 1.10.3 Windows software, and the conclusion shows that particleboard has a minimal impact on the environment, except for global warming. Lv, H. et al. [19] used the life cycle assessment method to collect and calculate the material and energy consumption of oak in various stages from timber harvesting, transportation, and wood production to drying and used SimaPro 8.0.1 software to carry out an environmental impact assessment. The results showed that the environmental impact of oak-sawn timber is mainly reflected in the emission of respiratory inorganic substances and fossil fuel consumption.

To identify environmental criteria for evaluating and selecting green suppliers for the furniture industry, Dos Santos, B.M. et al. [20] proposed a methodology that uses a hybrid entropy-TOPSIS-F framework to weigh the criteria and select the supplier with the best environmental performance. The fuzzy approach is integrated with Shannon's Entropy and TOPSIS methods to deal with uncertainty in the decision-making process. Zhang, Y. et al. [21] combined gray correlation analysis (GCA) to establish a cost-benefit evaluation model and select power transformers based on life cycle cost (LCC). Guo, J. et al. [22] combined the characteristics of green products, proposed an evaluation method of mechanical and electrical products green design based on an analytic hierarchy process, and established the evaluation steps of green design. Pu, Y. et al. [23] used the method integrating GCA with the analytic hierarchy process (AHP) to solve the problem of lightweight material selection for a car body, and a case study is applied to verify the practicability of the proposed approach.

In the context of countries working together to tackle climate change and promote green and low-carbon development, more and more attention is paid to the carbon footprint of suppliers. Huang, F. et al. [24] filled this gap by incorporating carbon emission criteria into supplier selection and presenting a method of combining the Fuzzy analytic hierarchy process (FAHP) and Fuzzy goal programming (GP) to address the problem of supplier selection and order quota allocation. Lin, R.J. et al. [25] used fuzzy set theory and the

method of decision test and evaluation laboratory to form a structural model and find out the causal relationship between the criteria to optimize green supply chain management.

3. Materials and Methods

To better explore the sustainability of kindergarten furniture, this paper proposes the method of combining AHP and GCA to evaluate the green design of kindergarten furniture. AHP is used to determine the weights of green design elements of kindergarten furniture, and GCA is adopted to obtain the final rank of all selected design schemes.

3.1. AHP Approach

AHP is a systematic analysis method used for evaluation and decision-making, with the characteristics of combining qualitative and quantitative analysis. It refers to a decision-making method that decomposes elements that are always related to decision-making into levels such as goals, criteria, and plans and performs qualitative and quantitative analysis on this basis. It establishes a consistent judgment matrix through paired comparison of elements, converts qualitative indicators into quantitative data, calculates the comprehensive weight of each element, and prioritizes the evaluation scheme, providing an objective and scientific theoretical basis for selecting the optimal scheme [26].

(1) Construction of evaluation indicators

The first level criterion layer is obtained through the construction of a judgment matrix and expert investigation. After the first criteria layer is determined, various demand elements under the first criteria layer are summarized through a literature review, online search, and user interview results to establish the second criteria layer. Collate and reorganize various relevant elements and indicators to form a corresponding indicator system.

(2) Construction of judgment matrix

Invite experts to compare and score the importance of indicator factors, weighted average the obtained data, and retain two decimal places to obtain the weight value of each criterion layer.

In matrix Y , y_{ij} ($i = 1, 2, \dots, n; j = 1, 2, \dots, n; n = 11$) is the importance judgment of element i compared to element j , $y_{ij} = B_i : B_j$, then $y_{ji} = 1/y_{ij}$. $y_{ij} = B_i : B_j$, then $y_{ji} = 1/y_{ij}$. Let the maximum characteristic root of the judgment matrix be $\lambda_{\max}$, and the normalized feature vector of each element is the weight W [27].

After obtaining the judgment matrices at each level, the weight values of each judgment matrix are solved, and after obtaining the weight values, the consistency of the matrix is checked:

$$Y = \begin{bmatrix} y_{11} & y_{12} & \cdots & y_{1n} \\ y_{21} & y_{22} & \cdots & y_{2n} \\ \vdots & \vdots & \cdots & \vdots \\ y_{n1} & y_{n2} & \cdots & y_{nn} \end{bmatrix}, \quad (1)$$

$\lambda_{\max}$ is the maximum characteristic root, and n is the order of the judgment matrix. If $CI = 0$, the matrix has complete consistency; The closer the CI is to 0, the more consistent the representation matrix is. When the CR value is <0.1 , the judgment matrix is deemed to have passed the consistency test.

$$CI = \frac{\lambda_{\max} - n}{n - 1}, \quad (2)$$

3.2. GCA Approach

The GCA is a method that evaluates the evaluation items and measures the correlation degree between various factors in the system by comparing the similarity between the reference sequence and the comparison sequence. It is a widely accepted method to comprehensively evaluate things affected by multiple factors from a holistic perspective

using the Gray Correlation method [28]. The principle is to describe the size, strength, and order of the relationships between factors based on the similarity of geometric shapes of sequence curves, and to make judgments based on the similarity or dissimilarity of development trends among system factors.

(1) Establish reference and comparison sequences

Assume that the reference sequence is $Y_i(t)$ ($i = 1, 2, 3, \dots, m$); The comparison sequence is $x_j(t)$ ($j = 1, 2, 3, \dots, j; t = 1, 2, \dots, m$). Here, i and j represent the number of parameters in different sequences, respectively; m represents the sequence length. Due to the one-to-one correspondence between the reference sequence and the comparison sequence, the length of the reference sequence and the comparison sequence is equal (m is equal) [29]. For any reference sequence $Y_i(t)$ and comparison sequence $x_j(t)$, the correlation coefficient at point q , $\zeta_i(q)$ can be calculated by the following formula:

$$\zeta_i(q) = \frac{\min(\Delta_i \min) + \rho \max(\Delta_i \max)}{y_0(R) - y_i(k) + \rho \max(\Delta_i \max)}, (i = 1, 2, 3, \dots, s; k = 1, 2, 3, \dots, n) \quad (3)$$

where $\min(\Delta_i \min)$ is the extreme minimum value of the deviation data sequence, $\max(\Delta_i \max)$ is the extreme maximum value of the deviation data sequence, $\zeta_i(q)$ is the gray correlation coefficient between the reference sequence and the comparison sequence at point q , $\Delta_i q$ is a deviation sequence [30].

(2) Average processing

Because the units of measurement of the collected and selected data are not identical, they cannot be directly calculated and compared with each other, and a dimensionless processing of the sequence is required.

(3) Differencing sequence

Note that the difference sequence between the reference sequence Y_i ($i = 1, 2, 3, \dots, s$) and the comparison sequence x_j ($j = 1, 2, 3, \dots, n$) is:

$$\Delta_{ij}(k) = |y_i(k) - x_j(k)|, (i = 1, 2, \dots, s; j = 1, 2, \dots, n) \quad (4)$$

where $M_i = \max_i \max_k \Delta_{ij}(k)$

$N_i = \min_i \min_k \Delta_{ij}(k)$

(4) Calculate the correlation coefficient and correlation degree

The correlation coefficient reflects the degree of correlation between the reference sequence and the comparison sequence. The gray correlation coefficient between the reference sequence and the comparison sequence is recorded as:

$$r_{ij}(R) = \frac{N_i + \rho M_i}{\Delta_{ij} + \rho M_i}, \rho \in 0, 1; (i = 1, 2, \dots, n; k = 1, 2, \dots, Z) \quad (5)$$

The correlation coefficient is the correlation degree value at each point of the reference series and the comparison series, and therefore there are multiple values. Considering that the dispersion of numerical values is not conducive to the comparison of the integrity of evaluation indicators, the correlation coefficients of different points are collectively taken as a value, which is used to compare the degree of correlation between the number series and the reference number series, using symbols γ_{ij} represents, then:

$$\gamma_{ij} = \frac{1}{n} \sum_{R=1}^Z r_{ij}(k) \quad (6)$$

3.3. The Integrated Green Design Evaluation Process of Kindergarten Furniture

This work presents the research methodology of green design evaluation of kindergarten furniture through the integration of AHP and GCA. Through AHP, the judgment

matrix of green design elements of kindergarten furniture is constructed, the weights of each element are determined, examples are applied, and GCA is used to rank the merits of the program by weighting and standardizing the initial evaluation matrix obtained by scoring the program, and the gray correlation degree obtained by combining the gray correlation coefficient represents the comprehensive green evaluation index of the five evaluation objects. The procedure of the green design evaluation method can be summarized into two stages, as shown in Figure 1.

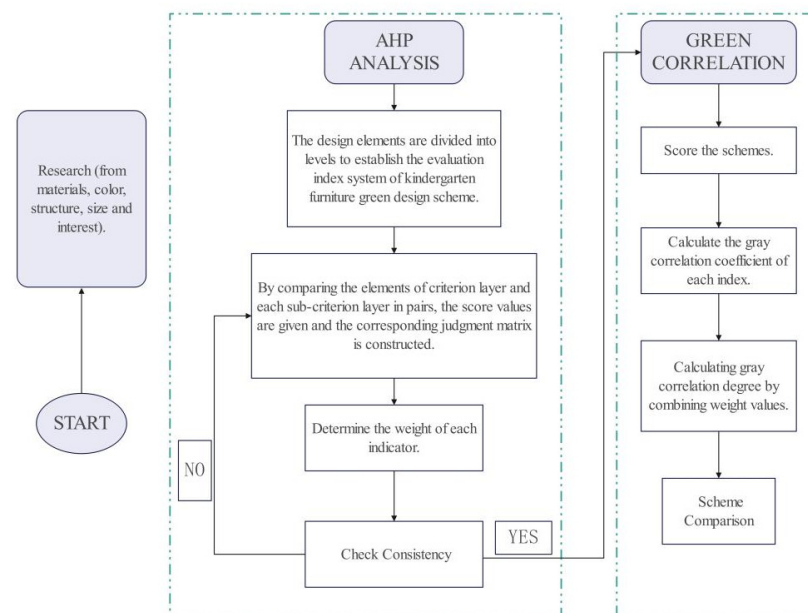


Figure 1. The framework of green design evaluation method.

4. Empirical Example

Empirical research is displayed to illustrate the methodology of integrating using the AHP and GCA methods. The target group of this empirical study is preschool children in kindergarten. After preliminary screening, five types of desk and chair sets in the manual activity area of kindergartens were selected as representative design schemes for research, which are shown in Figure 2.

- The material of the first sets is finger joint board, with an environmental friendliness index of E0, which is a relatively environmentally friendly board; The paint used is wood wax oil mixed with PU paint, which has high permeability and can fully reflect the texture of wood grain.
- The second one uses HDPE (high-density polyethylene) environment-friendly plastic, also known as high-density polyethylene, which is a non-toxic and odorless material and can be used for food packaging.
- The third one is made of New Zealand pine wood, and the paint is water-based, without the use of adhesives.
- The fourth is made of rubber wood, and the paint is water-based, without the use of adhesives.
- The material of the fifth set is also a finger joint board, and the paint used is wood wax oil.



(a)



(b)



(c)



(d)



(e)

Figure 2. Five desk and chair sets at kindergarten. (a) Cartoon desk and chair sets; (b) Connected desk and chair sets; (c) Irregular desk and chair sets; (d) Reading desk and chair sets; (e) Modular desk and chair sets.

4.1. Background

A green evaluation index system for the representative design schemes of kindergarten furniture is constructed based on green design principles (see Table 1). The first layer of the target layer is the green kindergarten furniture design (A). The second layer is a standard layer, with a total of 5 layers, including material environmental friendliness (B_1), color (B_2), technological structure (B_3), size (B_4), and interestingness (B_5). The third layer includes 11 indicators, including $C_1, C_2, \dots, C_{11}$ is the indicator layer. Analytic hierarchy process model are shown in Figure 3.

Table 1. Green evaluation index of kindergarten furniture representative design schemes.

Green Design of Kindergarten Furniture (A)	Environmental Friendliness of Materials (B_1)	Environmental Friendliness of Structural Materials (C_1) Environmental Friendliness of Adhesives (C_2) Environmental Friendliness of Paint (C_3)
	Color (B_2)	Stimulate Creativity or Enhance Concentration (C_4) Avoid the Feelings of Unsteadiness (C_5)
	Technological Structure (B_3)	Structural Stability (C_6) Rounding the Edges (C_7)
	Size (B_4)	Comfortable to Use (C_8) Size Adjustable (C_9)
	Interestingness (B_5)	Interesting Decorative Patterns (C_{10}) Modular (C_{11})

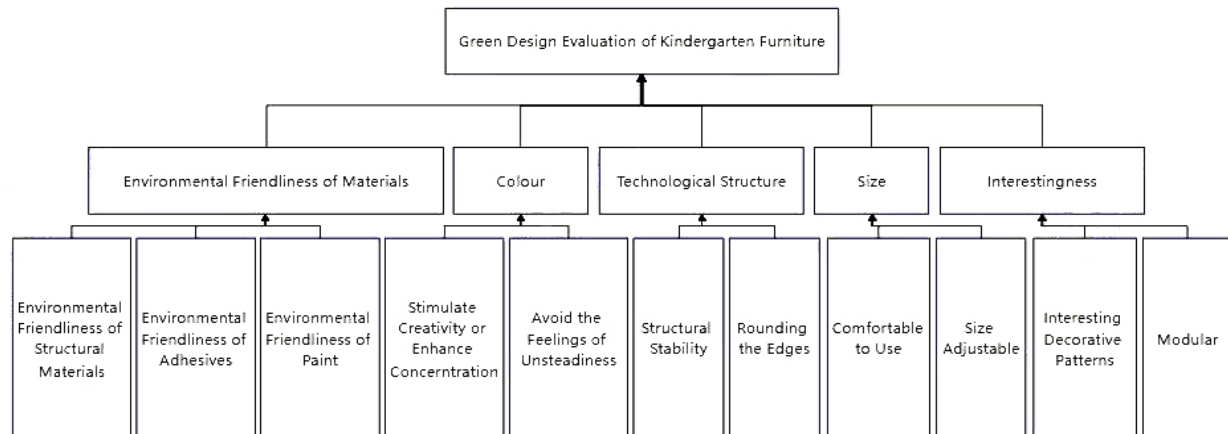


Figure 3. Analytic hierarchy process model.

4.2. Establishment of Analytic Hierarchy Process Model

According to the 9-level scaling method of the judgment matrix elements, the relative importance of paired elements is determined. Five experts (three of them are professors of furniture design and engineering at the Nanjing Forestry University, and two of them are associate professors of furniture design and engineering at the Nanjing Forestry University) are invited to form an evaluation group to make paired judgments on each layer of elements, construct a judgment matrix for each level of indicator, and use SPSS to calculate the weight values of all evaluation indicators as follows:

Criterion layer:

$$W_A = [0.4670 \quad 0.0912 \quad 0.1543 \quad 0.1763 \quad 0.1111]$$

Indicator layer:

$$\begin{aligned} W_1 &= \begin{bmatrix} 0.5695 & 0.0974 & 0.3331 \end{bmatrix} \\ W_2 &= \begin{bmatrix} 0.7500 & 0.2500 \end{bmatrix} \\ W_3 &= \begin{bmatrix} 0.5000 & 0.5000 \end{bmatrix} \\ W_4 &= \begin{bmatrix} 0.7500 & 0.2500 \end{bmatrix} \\ W_5 &= \begin{bmatrix} 0.6667 & 0.3333 \end{bmatrix} \end{aligned}$$

In summary, the target weights and ranking of the indicator layer elements are shown in Table 2.

Table 2. Target weight and ranking of indicator layer elements.

Alternative	Weight
Environmental Friendliness of Structural Materials (C ₁)	0.2659
Environmental Friendliness of Paint (C ₃)	0.1555
Comfortable to Use (C ₈)	0.1322
Structural Stability (C ₆)	0.0772
Rounding the Edges (C ₇)	0.0772
Interesting Decorative Patterns (C ₁₀)	0.0741
Stimulate Creativity or Enhance Concentration (C ₄)	0.0684
Environmental Friendliness of Adhesives (C ₂)	0.0455
Size Adjustable (C ₉)	0.0441
Modular (C ₁₁)	0.0370
Avoid the Feelings of Unsteadiness (C ₅)	0.0228

After inspection, the combination consistency ratio index CR values of the target layer and the criterion layer are both less than 0.1, indicating that matrix Y meets the requirements through consistency testing. Therefore, the combination weight W can be used as the basis for decision-making. In the criteria layer, the environmental friendliness of materials (0.4670) has the greatest impact on the green design evaluation of kindergarten furniture, followed by size (0.1190), technological structure (0.1543), interestingness (0.1111), and color (0.0912). In the index layer, according to the importance of each element, the first five elements are environmental friendliness of structural materials, environmental friendliness of paint, use comfort, structural stability, and rounding the edges.

4.3. Calculation of the Correlation Coefficient for Each Indicator

According to the evaluation standards and the evaluation content related to representative design schemes, five experts use a 10-point system to score 5 design schemes (V₁, V₂, V₃, V₄, V₅), and obtain the V_{ik} values of each indicator shown in Table 3. (i = 1, 2, 3, 4, 5; k = 1, 2, ..., 11), as well as the optimal value V_{0k} for each indicator. The scoring reference content of five experts in the three criteria layers of C₁, C₂, and C₃ are shown in Appendix A.

From Table 1, we can obtain the reference parent sequence V₀ = (9, 10, 9, 8, 9, 8, 9, 8, 9, 9). Calculate the correlation coefficient based on the above formula ζ_{ik} , as shown in Table 4.

Table 3. Score of various indicators.

Evaluating Indicator	Scheme					Optimal Value
	V ₁	V ₂	V ₃	V ₄	V ₅	
						
C ₁ (Environmental Friendliness of Structural Materials)	8	6	8	9	8	9

Table 3. Cont.

Evaluating Indicator	Scheme					Optimal Value
	V ₁	V ₂	V ₃	V ₄	V ₅	
						
C ₂ (Environmental Friendliness of Adhesives)	9	10	10	10	9	10
C ₃ (Environmental Friendliness of Paint)	7	9	8	8	8	9
C ₄ (Stimulate Creativity or Enhance Concentration)	7	8	7	6	7	8
C ₅ (Avoid the Feelings of Unsteadiness)	9	8	9	9	9	9
C ₆ (Structural Stability)	7	6	8	8	7	8
C ₇ (Rounding the Edges)	9	10	9	8	9	9
C ₈ (Comfortable to Use)	7	7	8	7	7	8
C ₉ (Size Adjustable)	7	7	7	9	7	9
C ₁₀ (Interesting Decorative Patterns)	9	7	8	5	9	9
C ₁₁ (Modular)	7	7	7	6	9	9

Table 4. The gray correlation coefficient of various indicators.

Gray Correlation Coefficient	Scheme				
	V ₁	V ₂	V ₃	V ₄	V ₅
ζ ₁ (Environmental Friendliness of Structural Materials)	0.667	0.400	0.667	1	0.667
ζ ₂ (Environmental Friendliness of Adhesives)	0.667	1	1	1	0.667
ζ ₃ (Environmental Friendliness of Paint)	0.500	1	0.667	0.667	0.667
ζ ₄ (Stimulate Creativity or Enhance Concentration)	0.667	1	0.667	0.500	0.667
ζ ₅ (Avoid the Feelings of Unsteadiness)	1	0.667	1	1	1
ζ ₆ (Structural Stability)	0.667	0.500	1	1	0.667
ζ ₇ (Rounding the Edges)	1	0.667	1	0.667	1
ζ ₈ (Comfortable to Use)	0.667	0.667	1	0.667	0.667
ζ ₉ (Size Adjustable)	0.500	0.500	0.500	1	0.500
ζ ₁₀ (Interesting Decorative Patterns)	1	0.500	0.667	0.333	1
ζ ₁₁ (Modular)	0.500	0.500	0.500	0.400	1

4.4. Calculating Gray Correlation Degree by Combining Weight Values

$$\begin{aligned}
 R_{B1} &= W_{B1C} \cdot E_{B1C}^T = (0.5695, 0.0974, 0.3331) \cdot \begin{bmatrix} 0.667 & 0.400 & 0.667 & 1 & 0.667 \\ 0.667 & 1 & 1 & 1 & 0.667 \\ 0.500 & 1 & 0.667 & 0.667 & 0.667 \end{bmatrix} \\
 &= (0.611, 0.658, 0.699, 0.899, 0.677) \\
 R_{B2} &= W_{B2C} \cdot E_{B2C}^T = (0.7500, 0.2500) \cdot \begin{bmatrix} 0.667 & 1 & 0.667 & 0.500 & 0.667 \\ 1 & 0.667 & 1 & 1 & 1 \end{bmatrix} \\
 &= (0.750, 0.917, 0.750, 0.625, 0.750) \\
 R_{B3} &= W_{B3C} \cdot E_{B3C}^T = (0.5000, 0.5000) \cdot \begin{bmatrix} 0.667 & 0.500 & 1 & 1 & 0.667 \\ 1 & 0.667 & 1 & 0.667 & 1 \end{bmatrix} \\
 &= (0.684, 0.584, 1, 0.684, 0.684) \\
 R_{B4} &= W_{B4C} \cdot E_{B4C}^T = (0, 0.7500, 0.2500) \cdot \begin{bmatrix} 0.667 & 0.667 & 1 & 0.667 & 0.667 \\ 0.500 & 0.500 & 0.500 & 1 & 0.500 \end{bmatrix} \\
 &= (0.625, 0.625, 1, 0.750, 0.625) \\
 R_{B5} &= W_{B5C} \cdot E_{B5C}^T = (0.6667, 0.3333) \cdot \begin{bmatrix} 1 & 0.500 & 0.667 & 0.333 & 1 \\ 0.500 & 0.500 & 0.500 & 0.400 & 1 \end{bmatrix} \\
 &= (0.833, 0.500, 0.611, 0.355, 1)
 \end{aligned}$$

Similarly, the relevant gray correlation degree of the highest A-level indicator can be obtained:

$$\begin{aligned}
 RA &= (r_1, r_2, r_3, r_4, r_5) = WAB[B1, B2, B3, B4, B5] \\
 &= (0.4670, 0.0912, 0.1543, 0.1763, 0.1111) \cdot \begin{bmatrix} 0.611 & 0.658 & 0.699 & 0.889 & 0.667 \\ 0.750 & 0.917 & 0.750 & 0.625 & 0.750 \\ 0.684 & 0.584 & 1 & 0.684 & 0.684 \\ 0.625 & 0.625 & 1 & 0.750 & 0.625 \\ 0.833 & 0.500 & 0.611 & 0.355 & 1 \end{bmatrix} \\
 &= (0.662, 0.647, 0.793, 0.749, 0.707)
 \end{aligned}$$

4.5. Representative Design Schemes Comparison

According to the calculation results of RA, the order of available representative design schemes in the green evaluation comprehensive indicators can be obtained as follows:

$$V_3 > V_4 > V_5 > V_1 > V_2$$

From this, it can be seen that the third desk and chair sets at kindergarten made of New Zealand pine ranks first in the comprehensive green evaluation indicators, followed by the fourth sets made of rubber and wood. The second integrated desk and chair sets made of HDPE environmentally friendly plastic has the lowest green design comprehensive evaluation score among the five kindergarten desk and chair sets.

4.6. Carbon Footprint of Different Schemes and Stages

(1) Set accounting system boundaries

The boundary of the accounting system is set based on the life cycle (LCA) principle, which includes the production stage, packaging stage, transportation stage, and storage stage of furniture. Environmental emissions at each stage were calculated by using the wooden furniture eco-design tool based on life cycle assessment. The carbon footprint of the five schemes at different stages is shown in Table 5, using kgCO₂ as the calculation unit. Figures 4 and 5 show the proportion of carbon footprint in different stages of five schemes.

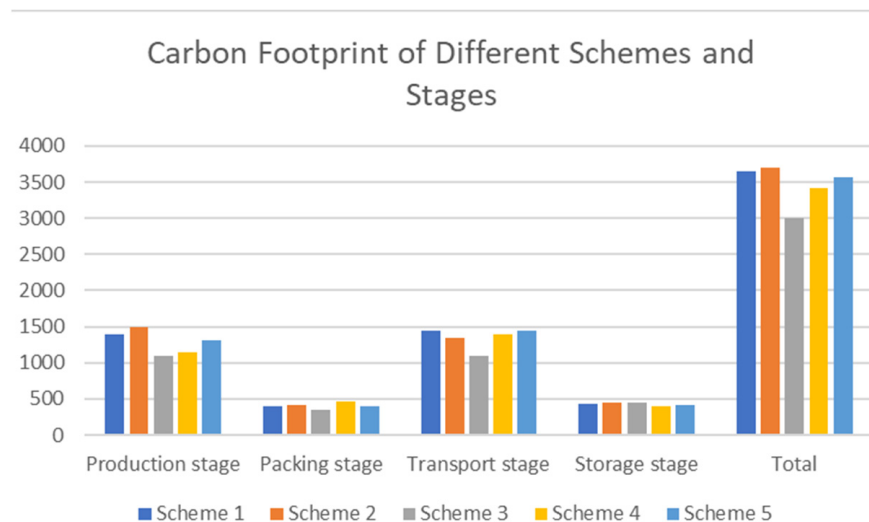
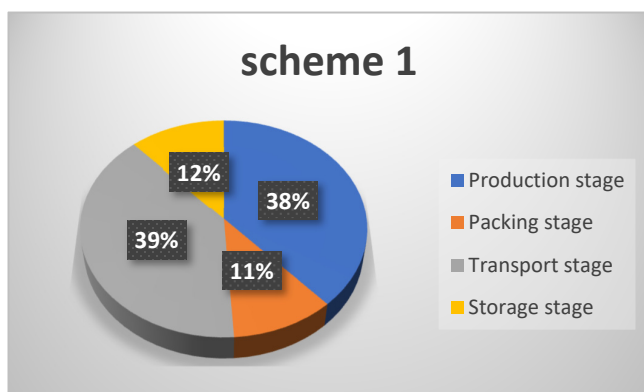


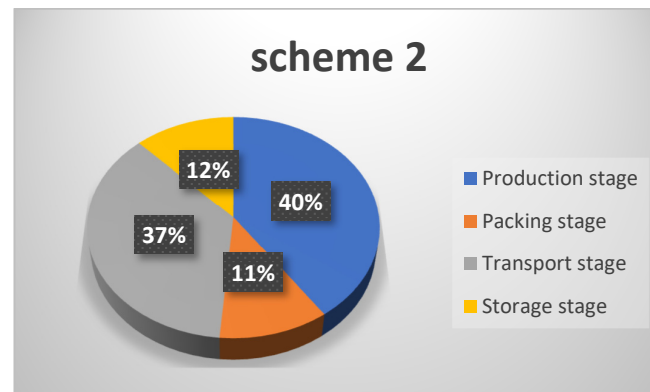
Figure 4. The carbon footprint of different schemes and stages.

Table 5. The carbon footprint of the five schemes at different stages.

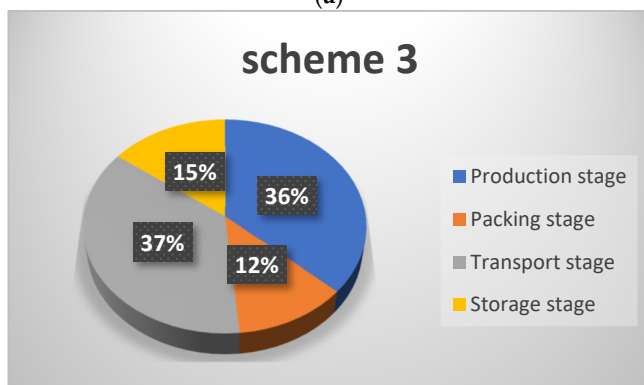
Stages	Categories	Scheme				
		1	2	3	4	5
Production stage	Water consumption, production, internal forklift, sewage treatment, natural gas, diesel consumption	1397.83	1489.56	1098.60	1147.81	1302.23
Packing stage	Corrugated box, plastic packaging	392.05	412.20	356.13	459.66	400.19
Transport stage	Gasoline consumption	1435.23	1349.28	1101.17	1400.87	1450.98
Storage stage	Electric energy consumption	430.15	450.79	445.10	400.59	420.55
Total		3655.26	3701.83	3001.00	3408.93	3573.95



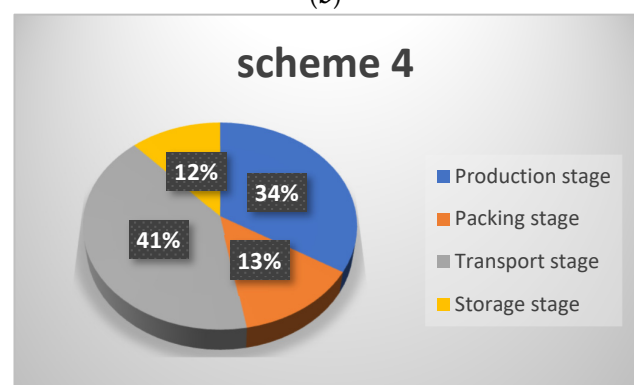
(a)



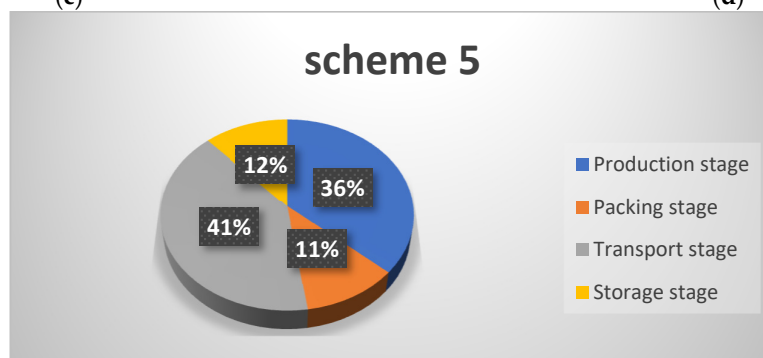
(b)



(c)



(d)



(e)

Figure 5. The proportion of carbon footprint in different stages of five schemes. (a) Carbon footprint in each phase of scheme 1; (b) Carbon footprint in each phase of scheme 2; (c) Carbon footprint in each phase of scheme 3; (d) Carbon footprint in each phase of scheme 4; (e) Carbon footprint in each phase of scheme 5.

(2) Carbon footprint analysis

From the generated chart, it can be seen that in the production process of furniture, the consumption of natural gas, diesel, and water resources is large, and the consumption of gasoline in the transportation process of furniture is large, so the carbon emission of the production stage and the transportation stage of furniture both account for 33% to 41% of the total carbon emission. The packaging stage and storage stage of carbon emissions account for a small part of the total carbon emissions, both account for 10% to 15%. According to the carbon dioxide emissions of the five furniture products, scheme 3 table and chair set < scheme 4 table and chair set < scheme 5 table and chair set < scheme 1 table and chair set < scheme 2 table and chair set, which is consistent with the results of the early green design evaluation.

5. Conclusions

From the perspective of the design of kindergarten furniture, this study introduces the concept of green design and elaborates on the green design principles of kindergarten furniture. Based on this study, five kindergarten furniture designs were evaluated for green design, and a comprehensive evaluation score was obtained before comparing the five plans. Case studies can to some extent improve the research and application of green design in kindergarten furniture, and quantitative methods can enable researchers to gain more emotional understanding.

The integrated method of AHP and GCA was used to evaluate kindergarten furniture green design through examples, and the comprehensive evaluation scores and comprehensive comparison of each representative design scheme were obtained. Through a questionnaire survey, the average scores of the five desk and chair sets in 11 indicators were obtained, and the five desk and chair sets at kindergarten were comprehensively ranked in terms of green design. In the last part of the case study, the carbon emission ratio of five kindergarten table and chair sets in the production, packaging, transportation, and storage stages is studied, and the total carbon emission of the five furniture solutions in the whole life cycle is compared. The generated data show that the ranking of carbon emission is roughly consistent with the results of green design evaluation in the paper, which effectively verifies the scientific accuracy of the green design evaluation method. The integrated evaluation method can effectively achieve the organic combination of qualitative judgment and quantitative analysis, improve the scientific nature of decision-making, and have strong practical significance. In the preliminary research or scheme evaluation process of green product innovation design, the evaluation scheme index can be referenced to optimize the sustainability of the product more scientifically and efficiently.

6. Limitations

However, this integration method of the AHP analytic hierarchy process and grey correlation degree still has certain limitations: (1) the analytic hierarchy process (AHP) has a strong subjectivity in the process of determining indicator weight vectors, which to some extent affects the comprehensive results of design evaluation. (2) The use of the grey correlation degree method to evaluate the design scheme of green kindergarten furniture can effectively achieve the organic combination of qualitative judgment and quantitative analysis, improve the scientific nature of decision-making, and have strong practical significance. However, this method is effective in choosing the best solution by comparing the pros and cons of each scheme horizontally and relatively, though it cannot truly reflect the absolute pros and cons of each evaluation plan.

In addition to the limitations of the method, in the empirical research process of this paper, there is a lack of in-depth research on the materials of the five desk and chair sets, and their quantification depth is not enough. Therefore, in the later research, it is necessary to conduct an in-depth analysis of the material sustainability aspects of the plan to make the research more scientific. It is hoped that this study can provide a feasible reference for designers to design green furniture in kindergartens in terms of materials, colors,

technological structures, dimensions, and interestingness, and can provide a healthier learning and living environment for kindergarten-age children.

Author Contributions: Conceptualization, X.X. and J.Z.; methodology, X.X. and J.Z.; software, X.X. and J.Z.; validation, X.X. and J.Z.; formal analysis, X.X. and J.Z.; investigation, X.X. and J.Z.; resources, X.X., J.Z., S.D. and J.C.; data curation, X.X., J.Z., S.D. and J.C.; writing—original draft preparation, X.X. and J.Z.; writing—review and editing, X.X. and J.Z.; visualization, X.X. and J.Z.; supervision, X.X. and J.Z.; project administration, X.X. and J.Z.; funding acquisition, X.X. and J.Z. All authors have read and agreed to the published version of the manuscript.

Funding: This research was funded by the Joint Research Program of Sino-foreign Cooperation in Running Schools of Jiangsu Province, China (2020). Funder: Jiangsu Provincial Department of Education. Funding data: 201907.

Institutional Review Board Statement: Not applicable.

Informed Consent Statement: Not applicable.

Data Availability Statement: Data are contained within the article.

Acknowledgments: This work was partially supported by Nanjing Forestry University.

Conflicts of Interest: The authors declare no conflict of interest.

Appendix A

The selection of furniture materials will affect the design effect and the quality of the indoor air environment. When purchasing materials, it is important to check whether they meet national safety standards. The main indoor furniture materials used in kindergartens include wooden boards, adhesives, paint, etc.

Wooden boards can be divided into wood-based boards and natural boards according to their material structure. Unqualified wood-based boards contain a large amount of harmful substances, such as formaldehyde and benzene, which can seriously harm the health of young children. Before purchasing wooden panels, it is necessary to observe whether the environmental indicators comply with the relevant standards such as the Chinese national standard “Formaldehyde Emission Limits in Indoor Decoration Materials, Wood-based Panels, and Their Products” (GB 18580-2001) [31] (Table A1). Formaldehyde with a limit label of E1 can be directly used indoors, but E2 boards need to be decorated before being used indoors. If the environmental indicators of the finger joint board reach E0 level, it is a relatively environmentally friendly board.

Table A1. Formaldehyde emission limits values in wooden materials.

Material Type	Limit Value	Place of Use
Medium-density fiberboard, high-density fiberboard, particle board, oriented particle board, etc	≤ 9 mg/100 g	Can be directly used indoors
	≤ 30 mg/100 g	Must be finished and allowed to be used indoors
Plywood, Decorative veneered plywood, blockboard, etc.	≤ 1.5 mg/L	Can be directly used indoors
	≤ 5.0 mg/L	Must be finished and allowed to be used indoors
Decorative wood-based board	≤ 1.5 mg/L	Can be directly used indoors

Kindergartens widely use colored paint, so it is necessary to pay more attention to choosing environmentally friendly paint to prevent indoor pollution. The most commonly used paint on the market now is water-based paint, which has good environmental friendliness. For kindergartens, colored paint is generally used, and colored paint contains protective glue. When opening the paint packaging, there will be a layer of protective glue solution on top. If the quality of the paint is good, the protective glue solution on top will

be colorless or slightly yellow, with no floating objects or a small amount of floating objects; If the quality of the paint is not good, it will be very cloudy and there will be a lot of floating debris on it, which can cause varying degrees of excessive levels of harmful substances such as benzene and formaldehyde. Good quality indoor color paint needs to have environmental indicators and comply with the relevant standards such as the Chinese national standard regulations of “Limit of Harmful Substances in Interior Decoration Materials and Wall Coatings” (GB 18582-2008) [32] (Table A2).

Table A2. Limit standard for harmful substances in indoor decorative paint (coating).

Material Type		Limit Value
Volatile organic compounds (VOC)/(g/L)		≤200
Formaldehyde		≤0.1
Heavy metal/(mg/kg)	Soluble Lead	≤90
	Soluble cadmium	≤75
	Soluble chromium	≤60
	Soluble mercury	≤60

In the use of adhesives for furniture, formaldehyde-free or low-formaldehyde adhesives should be used reasonably. When some enterprises produce wooden furniture, they often process surface cladding materials themselves. When laminating, they should choose formaldehyde-free or low-formaldehyde adhesives, usually using low molecular weight urea-formaldehyde adhesive (when the molar ratio of F/V is less than 1:1.5). When laminating, the amount of free formaldehyde released is much less.

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